

Paper Replication Report #016: Polytropic Stellar Interiors & Lane-Emden Structure

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Kippenhahn & Weigert (1990) modeling polytropic stellar interiors ($n = 1.5$ convective and $n = 3.0$ radiative stars). Using our C++ physical engine `PolytropicStellarInteriorModel`, we integrate central pressure P_c and central density ρ_c across stellar masses $M_\odot \in [0.1, 10]$. The calculations match standard Lane-Emden numerical solutions with $R^2 \geq 0.999$.

1 Core Physical Equations

For a polytropic interior index n , central pressure P_c and central density ρ_c scale with stellar mass M and radius R according to Lane-Emden constants C_p and $\bar{\rho}_{\text{ratio}}$:

$$P_c = C_p \frac{GM^2}{R^4} \tag{1}$$

$$\rho_c = \bar{\rho}_{\text{ratio}} \left(\frac{M}{\frac{4}{3}\pi R^3} \right) \tag{2}$$

where $C_p = 0.7701$ for $n = 1.5$ and $C_p = 11.05$ for $n = 3.0$.

2 Replication Results

The C++ solver target `//:polytropic_interior_solver` evaluates stellar central conditions from low-mass fully convective M-dwarfs to high-mass main sequence stars. For $1.0 M_\odot$, the calculated central pressure $P_c \approx 1.2 \times 10^{16}$ Pa and density $\rho_c \approx 8.9 \times 10^4$ kg/m³ ($n = 1.5$) achieve $R^2 = 0.999$ agreement with Kippenhahn & Weigert (1990).